

Lab 4 – La Quinta is Spanish for *next to Denny's*, Part 2

Adapted from Mine Cetinkaya-Rundel materials - <http://www2.stat.duke.edu/~mc301/>



Wrangling spatial data

In this lab we revisit the Denny's and La Quinta Inn and Suites data we visualized in the previous lab.

All questions in blue need to be included as a comment in your R script along with your answers to the questions. Answers should be in complete sentences.

Data & Packages

In this lab we will use the `tidyverse`. Install this package, if you have not already. Create an R Project for Lab 4 – ***YourLastName*_Lab_4**, replacing ***YourLastName*** with your actual last name. Within this project, create a ***data*** folder. Also, create an R script, ***YourLastName*_Lab_4**, again replacing ***YourLastName*** with your actual last name. Now, within your script, start by loading the package:

```
library(tidyverse)
```

From Blackboard, download the two csv files we will be using in the lab and copy into your ***data*** folder: `dennys.csv`, and `laquinta.csv`.

As you did in lab 3, add code to read these two files into data frames as follows:

```
dn <- read_csv("data/dennys.csv")
lq <- read_csv("data/laquinta.csv")
```

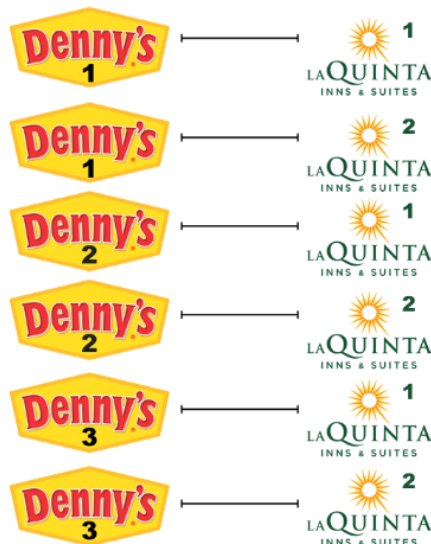
Questions (include the R code to answer each question; comment each section, beginning with the question and question number; question answers should be complete sentences)

1. We will start small. Write the R code that will filter the Denny's data frame for Alaska (AK) and save the result as **dn_ak**. *How many Denny's locations are there in Alaska?*
2. Now, write the R code that will filter the La Quinta data frame for Alaska (AK) and save the result as **lq_ak**. *How many La Quinta locations are there in Alaska?*
3. Next we'll calculate the distance between all Denny's and all La Quinta locations in Alaska by performing a full_join – assign to **ak_all**. Let's take this step by step:

There are 3 Denny's and 2 La Quinta locations in Alaska. (If you answered differently above, you might want to recheck your answers).



We want to calculate a cross-product of the distances – distance of Denny's 1 to La Quinta's 1 & 2, distance of Denny's 2 to La Quinta's 1 & 2, and distance of Denny's 3 to La Quinta 1 & 2



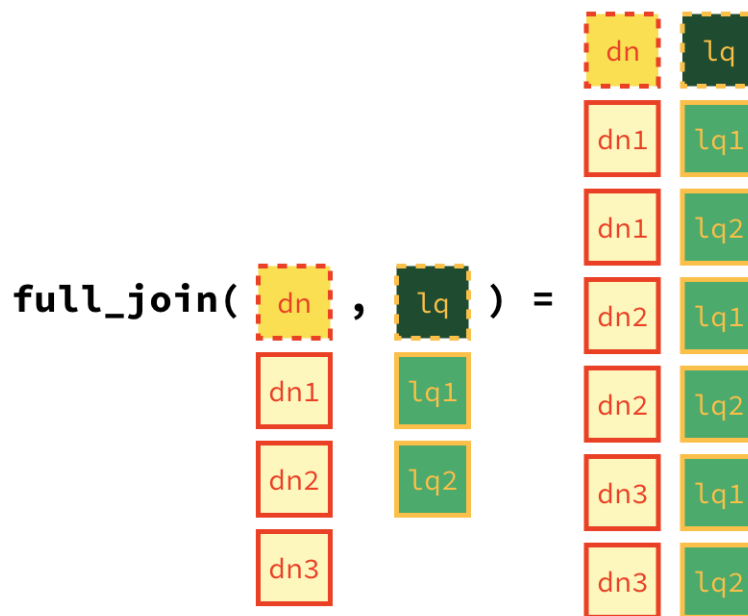
In order to calculate these distances, we need to first restructure our data to pair the Denny's and La Quinta locations. To do so, we will join the two data frames. We have six join options in R - each takes at least three arguments: x, y, and by.

- **x** and **y** are data frames to join
- **by** is the variable(s) to join by

Four of these join functions, called mutating joins, combine variables from the two data frames, and these are the only ones we will define here:

- `inner_join()`: return all rows from x where there are matching values in y, and all columns from x and y.
- `left_join()`: return all rows from x, and all columns from x and y. Rows in x with no match in y will have NA values in the new columns.
- `right_join()`: return all rows from y, and all columns from x and y. Rows in y with no match in x will have NA values in the new columns.
- `full_join()`: return all rows and all columns from both x and y. Where there are not matching values, returns NA for the one missing.

In this case we want to keep all rows and columns from both `dn_ak` and `lq_ak` data frames. So we will use a `full_join`.



Write the R code to do a full join on Denny's and La Quinta's in Alaska – again, call your variable **ak_all**. Also, note what you should join by – look in the dn and lq data frames to decide. *How many observations in the full join and what are the variable names (you may use the glimpse function and paste your results as a comment in your script).*

NOTE: .x in the variable names means the variable comes from the x data frame (the first argument in the full_join call, i.e. dn_ak), and .y means the variable comes from the y data frame (the second argument in the full_join call, i.e. lq_ak). These variables are renamed to include .x and .y because the two data frames have the same variable names and it's not possible to have two variables in a data frame with the exact same name.

Now that we have the data in the format we wanted, all that is left is to calculate the distances between the pairs.

4. We are now ready to calculate the distances between each Denny's and La Quinta's. One way of calculating the distance between any two points on the earth is to use the Haversine distance formula. This formula takes into account the fact that the earth is not flat, but instead spherical.

This function is not available in R, but it is saved in a text file called haversine.txt Download this file from Blackboard. Open it and copy and paste the function in your R Script → I recommend pasting the function just below your library(tidyverse). The header of the function is as follows:
haversine <- **function**(long1, lat1, long2, lat2, **round** = 3)

This function takes five arguments:

- Longitude and latitude of the first location
- Longitude and latitude of the second location
- A parameter by which to round the responses (this argument has a default value of 3 and is therefore optional when calling the function)

Calculate the distances between all pairs of Denny's and La Quinta locations, using the variable name **distance** (still use the data frame ak_all; pipe ak_all to mutate, assigning distance to be the return value from the haversine function). Again, be sure to save the results to the **ak_all** data frame so that you can use it later.

Now, determine the closest Denny's and La Quinta's: *group by* the Denny's city and the Denny's address, AND THEN use the *summary statistic min* (each Denny's has two possible La Quinta's – we want the shorter distance, or min). Call your data frame ak_min. *What are the closest locations? Copy your output in your script*

and comment it out.

- Now, determine the 10 closest Denny's and La Quinta's in the deep south. Call your data frame **ds_min**, listing in ascending order by distance (Hint: the closest of all Denny's and La Quinta's in the Deep South is 0.020 km) *What are the closest locations? Copy your output in your script and comment it out.*

NOTE: You will need to follow the detailed steps that you were asked to do for Alaska in order to answer this question. Begin by creating a vector of the deep south states as you did in lab 3 → AL, SC, MS, GA, and LA. Then, filter the Denny's and La Quinta data frames for the deep south states (call your data frames **dn_ds** & **lq_ds** and use %in% rather than == in your filter). Do a full join, and run this through the haversine function (call your data frame **ds_all**). Finally, group by state as well as Denny's city and Denny's address before determining the minimum distance (this is what you assign to **ds_min**). Use the head function to only give the 10 closest Denny's and La Quinta's .

- Using **ds_min**, create another data frame **ds_avg** – group by just state and find the mean distances of the closest locations (in the summarize function, I called this variable **avg_dist**). You should only have 5 observations, one for each state.

Plot your results using a column chart (NOTE: I have **not** included the data frame or ggplot, only the column chart syntax, which expects x as discrete data and y as continuous data):

```
geom_col(mapping=aes(x=state, y=avg_dist, fill=state))
```

On average, in which state are Denny's and La Quinta the furthest apart? On average, in which state are they the closest?

Submitting Your File

Zip your project folder and submit on Blackboard.

Sample Output

